

WHY PATH A FAILS: THE ROLE OF THE WEIL CONSTANT AND THE LIMITS OF EDGE-MASS ASYMPTOTICS

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ABSTRACT. We give a self-contained account of why the Path A sufficient condition $\tilde{q}_L^\theta := T + \theta V + K_L - c_L I \geq 0$ fails at $L = 7/20$ and $\theta = 69/100$. An independent 128-bit Arb computation clarifies the spectral mechanism: the negative direction is driven by the *Weil constant* $c_L \approx 1.365$ acting as a global negative shift, not by a competition between the Archimedean kernel and edge-mass absorption. As a byproduct, we identify a failure of the asymptotic $\mu_K^{(02)} \sim c_2/\kappa_{\text{edge}}(L)$ (companion Lemma L3) in the interior of the first-prime window, which invalidates the previously proposed formula $\theta_0(L) = 1 - c_2/\kappa_{\text{edge}}(L)$ for the off-diagonal flip point. All numerical claims are verified by the scripts in `src/verify_2x2_gram.py` and `checker/thm6/check_thm6.py`.

1. SETUP AND NOTATION

Let $L = 7/20$, $\theta = 69/100$, $c_L \approx 1.36527$ (Arb-certified to 256-bit precision from the Bombieri–Weil explicit formula; see [1] Remark 2.1). The Weil quadratic form is

$$Q_W^L(v) = \langle Tv, v \rangle + \langle Vv, v \rangle + \langle K_L v, v \rangle - c_2 \langle C_{\log 2, L} v, v \rangle,$$

where $V(x) = -\frac{1}{2} \log(1 - x^2)$, K_L has kernel $k_L(x, y) = -L r''(L(x - y))$, $C_{\log 2, L}$ is the prime-2 convolution operator, and $c_2 = \log 2/\sqrt{2}$.

The Path A sufficient condition for $Q_W^L \geq 0$ replaces the prime term by absorbing it into θV :

$$\tilde{q}_L^\theta := T + \theta V + K_L - c_L I.$$

Theorem 6 of [1] shows this condition is false at $(L, \theta) = (7/20, 69/100)$ via two explicit negative witnesses. This note explains the underlying mechanism in terms of the 2×2 Gram matrix in the $\{P_0, P_2\}$ Legendre subspace, and corrects an error in an earlier draft that attributed the failure to edge-mass asymptotics.

2. THE 2X2 SPECTRAL MECHANISM

In the Suzuki framework [1], the Legendre basis $\{P_n\}$ is unnormalised with Gram matrix $G_{ij} = \frac{2}{2i+1} \delta_{ij}$. The kinetic term acts as $(T_N)_{ij} = H_{n_j} G_{ij}$ where $H_n = \sum_{k=1}^n k^{-1}$ is the n -th harmonic number. The Gram matrix entries of $\tilde{q}_L^\theta$ on $\{P_0, P_2\}$ are:

$$E_{mn}^\theta = H_{n_m} G_{mm} \delta_{mn} + \theta \langle V P_n, P_m \rangle + \langle K_L P_n, P_m \rangle - c_L G_{mm} \delta_{mn}.$$

Remark 2.1 (Numerical values at $L = 7/20$, $\theta = 69/100$). A 128-bit Arb computation (script `src/verify_2x2_gram.py`, parameter `a = L = 7/20`) gives:

Quantity	Value	Comment
$\langle VP_0, P_0 \rangle$	0.6137	$V > 0$ on I_L
$\langle K_L P_0, P_0 \rangle$	2.4890	large positive
$H_0 G_{00} = 0$	0	P_0 has zero kinetic energy
$c_L G_{00}$	$1.365 \times 2 = 2.730$	dominant negative shift
$E_{00}^{0.69}$	$\approx +0.163$	net: small positive
$\langle VP_2, P_2 \rangle$	0.2694	
$\langle K_L P_2, P_2 \rangle$	-0.00019	tiny
$H_2 G_{22} = 0.6$	0.6	kinetic contribution
$c_L G_{22}$	$1.365 \times 0.4 = 0.546$	
$E_{22}^{0.69}$	$\approx +0.222$	net: small positive
$\langle VP_0, P_2 \rangle$	0.3333	
$\langle K_L P_0, P_2 \rangle$	0.00712	
$E_{02}^{0.69}$	$\approx +0.218$	off-diagonal
$\det(E^\theta)$	≈ -0.011	negative

Theorem 2.2 (Spectral mechanism for Path A failure). *At $L = 7/20$, the 2×2 matrix E^θ has a strictly negative eigenvalue for all $\theta \in [0, \theta_{\max}]$ where $\theta_{\max} \approx 0.997$. The negative direction is $v^* = E_{22}P_0 - E_{02}P_2 \approx P_0 - P_2$.*

The negativity is not caused by the off-diagonal entry $E_{02}^2 > E_{00}E_{22}$, but by $c_L > 0$ reducing the diagonal entries below the level needed to dominate the off-diagonal. Without c_L (i.e., $c_L = 0$), the matrix is positive definite: $\det E^\theta|_{c_L=0} \approx +1.48 > 0$.

Proof. Direct computation from Remark 2.1: $\det E^\theta = E_{00}E_{22} - (E_{02})^2 \approx 0.163 \times 0.222 - 0.218^2 \approx 0.036 - 0.047 = -0.011 < 0$. Setting $c_L = 0$ removes the -2.730 from E_{00} and -0.546 from E_{22} , giving $E_{00}|_{c_L=0} \approx 2.893$ and $E_{22}|_{c_L=0} \approx 0.768$, hence $\det|_{c_L=0} \approx 2.893 \times 0.768 - 0.218^2 \approx 2.222 - 0.047 > 0$. $\square$

Corollary 2.3 (Why the Schur complement machine is necessary). *Because $\{P_0, P_2\}$ is positive definite when $c_L = 0$, a redistribution-only argument ($\tilde{q}^\theta$ without the c_L shift) cannot detect the obstruction. The full Split-residual Schur criterion of [1] Theorem 5 is necessary precisely because it must overcome the global negative shift $c_L I$ using the positive kinetic energy of high-order modes $H_n \gg c_L$ for large n .*

3. FAILURE OF THE EDGE-MASS ASYMPTOTIC

An earlier draft of this paper (now withdrawn) attempted to characterise the Path A obstruction via the formula $\theta_0(L) = 1 - c_2/\kappa_{\text{edge}}(L)$, claiming this equals the zero of the off-diagonal entry $E_{02}^\theta(L)$. This was incorrect.

Remark 3.1 (Lemma L3 fails in the physical window). The companion paper's Lemma L3 states $\langle K_L P_0, P_2 \rangle \sim \kappa_{\text{edge}}(L)^{-1} c_2$ as $\varepsilon \rightarrow 0^+$, where $\varepsilon = 2 - \log 2/L$. At $L = 7/20$, $\varepsilon \approx 0.020$ and the asymptotics give $c_2/\kappa_{\text{edge}} \approx 0.303$, while the certified value is $\langle K_L P_0, P_2 \rangle \approx 0.00712$. The $O(1)$ remainder (≈ -0.296) is the entire leading term; the asymptotic provides no useful bound in the first-prime window.

As a consequence, the actual zero of E_{02}^θ satisfies:

$$\theta_{\text{zero}} = \frac{c_2 \gamma_{02} - \langle K_L P_0, P_2 \rangle}{\langle V P_0, P_2 \rangle} \approx \frac{0.0186 - 0.00712}{0.3333} \approx 0.034,$$

far from $\theta_0 \approx 0.697$. The formula $1 - c_2/\kappa_{\text{edge}}$ has no algebraic relationship to the spectrum of $\tilde{q}^\theta$.

4. SUMMARY

- (1) Path A fails because the Weil constant $c_L \approx 1.365$ globally shifts the diagonal entries of the 2×2 Gram matrix below the level needed for positive definiteness. Without c_L , the $\{P_0, P_2\}$ subspace is positive definite.
- (2) Lemma L3's asymptotic $\mu_K^{(02)} \sim c_2/\kappa_{\text{edge}}$ fails by a factor of 40 at $L = 7/20$; any argument depending on it is invalid in the first-prime window.
- (3) The formula $\theta_0 = 1 - c_2/\kappa_{\text{edge}}$ is not the zero of E_{02}^θ . Theorem 6 of [1] provides the correct (certified) falsification of Path A without requiring any asymptotic expansion.

REFERENCES

- [1] Lin Tao, *Structural lemmas for the first-prime window of the Weil quadratic form*, preprint (2026). doi:10.5281/zenodo.21807498.